



Predicted measurements and innovation

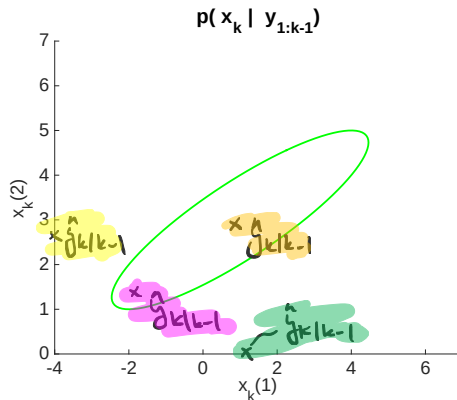
- In the update step of the Kalman filter we use the measurement $\mathbf{y}_k$ to compute the posterior density of the state $\mathbf{x}_k$.
- The posterior mean can be expressed as

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \mathbf{v}_k,$$

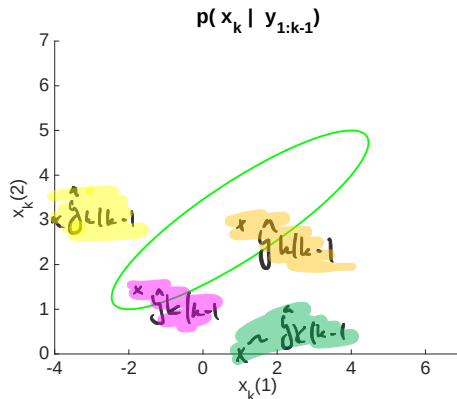
where $\mathbf{v}_k = (\mathbf{y}_k - \hat{\mathbf{y}}_{k|k-1})$ is the *innovation*.

- Does this make sense?
Let us visualize what $\hat{\mathbf{y}}_{k|k-1}$ and $\mathbf{v}_k$ represent in a few examples!
- **Example:** suppose $\mathbf{x}_k$ is a 2D-vector that contains position and velocity in 1D.

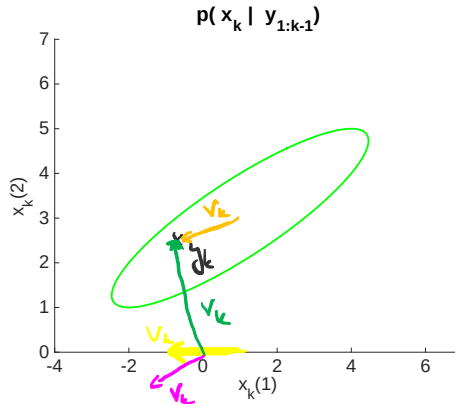
- Suppose $\mathbf{y}_k = \mathbf{x}_k + \mathbf{r}_k$ where $\mathbf{r}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$.
- Consider the following predicted density and your alternatives:



- Suppose $y_k = \mathbf{x}_k(1) + r_k$ where $r_k \sim \mathcal{N}(0, R_k)$.
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- Suppose $\mathbf{y}_k = \mathbf{x}_k + \mathbf{r}_k$ where $\mathbf{r}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$.
- Guess $\mathbf{v}_k$ for $p(\mathbf{x}_k | \mathbf{y}_{1:k-1})$ and $\mathbf{y}_k$ below



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